

Simpson's Paradox — Corpus Summary

A digital mirror of Simpson's Paradox — the paradox falls out of the DAG as a derived fact, never modeled directly.

Generated 2026-07-21 from live vw_* views — same data as /conclusions. SSoT is the rulebook JSON.

Scope boundary (conc-09)

This rulebook is a high-fidelity measurement instrument, not an oracle of truth. It classifies allocation geometry precisely and repeatably — sign flips, distortion ratios, signal purity — the same way a thermometer measures temperature. It does not adjudicate causal truth. Causal interpretation is external, gated by AdjustmentAppropriate, and rests on CausalRole, the one hand/LLM-encoded field in the DAG. Berkeley: highest AllocationDistortion (0.193) but CausalRole=contested — geometry and confounding are not the same thing, and the instrument only measures the former.

Scope & limits

Theorem conclusions follow from definitions (regression checks for the transpiler). Domain conclusions and corpus discovery findings are statistics on a curated convenience sample — directional, not inferential. See conc-14 before treating cross-domain patterns as laws.

Allocation sweep: 10 steps per study, f " [0.05, 0.95] (SweepStudyConfig). Latent flip = Type D at observed allocation + PooledGapCrossesZero somewhere in that range — not necessarily at observed allocation.

How to read "InvariantChecks" (read before the pass count)

This instrument is a measurement device, not an oracle: it classifies allocation geometry precisely and repeatably, the way a thermometer classifies temperature — it does not adjudicate whether a study's causal story is correct. "Invariant" in this rulebook means "the transpiler output is self-consistent with its own definitions," not "an independent empirical hypothesis survived a test." Three different things get called an invariant here, and conflating them is the single easiest way to overstate this corpus:

1. Definitional (cannot fail by construction). DistortionType, IsSignFlip, AllocationDirection, and SignalPurity are all calculated from the same three numbers (WeightedStratumGapSum, SignedPooledGap, CorrectedGap, AllocationDistortion). An invariant like "DistortionType " {A,B} ' !' IsSignFlip=TRUE" restates one algebraic fact in a second vocabulary — it is a regression check that the transpiler did not corrupt a formula, not a finding about the world. Most of the 32 rows in InvariantChecks are this kind.
2. Judgment-crossing (the closest thing to a substantive check). CausalRole on StratumVariables (confounder / mediator / collider / selection / contested / unknown) is the one hand/LLM-encoded field in the DAG — schema type "raw," not derived from geometry. Invariants that reference StratumCausalRole, ConfounderIdentity, or IsParadoxExplained (e.g. inv-explained-sign-flip-confounder, inv-collider-no-manifest) are checking that the geometric result is consistent with that human judgment call — the only place a real counterexample could show up. The rulebook does not currently record who made each CausalRole call or by what protocol; treat these classifications as reviewed labels, not as independently verified ground truth.
3. Vacuous (PassCount=0). A handful of invariants (e.g. inv-type-a-unanimous, inv-confounder-signflip-explained, inv-collider-no-manifest) currently have zero rows in their filtered universe. "0 pass / 0 fail" means no study in this corpus falls into that category yet — it is true the way "all unicorns are blue" is true, and proves nothing until a qualifying study exists.

Epistemic summary (tiered)

Proved by construction (pure algebra — no CausalRole dependency): 4
Proved, conditional on CausalRole annotation (see caveats below — not pure algebra): 4
Established in instrument (instrument · taxonomy · methodology · scope): 24
Observed in this corpus (domain — provisional): 2
Total conclusions: 34
Corpus hypotheses (loop-61): 20 / 22 confirmed
Vacuous (true by construction — excluded from ratio above): 3 / 3 pass
Consistency checks (definition-linked): 9 / 9 pass

Latent Type-D (under sweep contract above): 112 / 138 (81.2%)

Studies in corpus: 242

Distortion taxonomy: A 79 · B 10 · C+ 8 · C" 7 · D 138

Pre-registered discovery (loop-61)

Consistency checks — confirm implementation matches stated algebra (same tier as conc-12 / inv-signal-purity-sign-flip). Not independent discoveries about nature.

H-collider-no-manifest-theorem — PASS (consistency check)

At scale ("e10 collider/selection studies post expansion), zero manifest sign-flips: ColliderSelectionManifestCount = 0.

Expected: ColliderSelectionManifestCount = 0 AND ColliderSelectionCount >= 10

Observed: collN=11 collManifest=0

H-corrected-gap-invariant — PASS (consistency check)

CorrectedGap is allocation-invariant for every study: MAX(SweepStudySummary.SweepCorrectedGapRange) < 0.0001 and zero failing studies.

Expected: MaxStudySweepCorrectedGapRange < 0.0001 AND CorrectedGapInvariantFailCount = 0

Observed: maxRange=0.000 fails=0

H-explained-bidirectional — PASS (consistency check)

IsParadoxExplained 'ú confounder manifest sign-flip: ExplainedConfounderCount = ConfounderSignFlipCount, no false-positive explained rows, no unexplained confounder sign-flips.

Expected: ExplainedConfounderCount = ConfounderSignFlipCount AND FalsePositiveExplainedCount = 0 AND UnexplainedConfounderSignFlipCount = 0

Observed: explained=85 confFlip=85 falsePos=0 unexplained=0

H-identity-map-coverage — PASS (consistency check)

At least 95% of real-study StratumVariables carry a ConfounderIdentity map — ontology covers the corpus.

Expected: IdentityMapCoverageRate >= 0.95

Observed: IdentityMapCoverageRate=1.000

H-perfect-balance-sufficient-safety — PASS (consistency check)

Perfect allocation balance (ArmSizeRatio=0.5 AND MaxStratumImbalance=0.0) is a domain-agnostic sufficient condition for no manifest Simpson's Paradox — no such study produces IsSignFlip=TRUE regardless of mechanism class, domain, or confounder identity.

Expected: PerfectBalanceFlipCount=0 AND PerfectBalanceStudyCount>=100

Observed: PerfectBalanceFlipCount=0 PerfectBalanceStudyCount=118

H-purity — PASS (consistency check)

Every sign-flip study has SignalPurity strictly below 0.5.

Expected: SignFlipSignalPurityMax < 0.5

Observed: maxPurity=0.500

H-purity-inversion — PASS (consistency check)

Mechanism classes with PolicyDefault='do-not-condition' or 'do-not-naively-adjust' (collider-selection, selection-frailty) have strictly higher mean SignalPurity than all stratify-* mechanism classes. High SignalPurity anti-predicts safe stratification — it is a trap for researchers who use data cleanliness as a quality heuristic.

Expected: MeanPurityColliderSelection > MeanPurityAllStratifyClasses AND MeanPuritySelectionFrailty >= 0.95

Observed: MeanPurityDoNotCondition=0.908 MeanPurityStratifyClasses=0.728

H-theorem-portfolio — PASS (consistency check)

Four Category=theorem conclusions are witnessed: SignalPurity (conc-12/20) plus three new theorems from loops 73–75.

Expected: TheoremCount >= 4

Observed: theoremCount=8

H-unexplained-flips-only-nonconfounders — PASS (consistency check)

All unexplained sign-flips ($\text{IsSignFlip}=\text{TRUE}$ AND $\text{IsParadoxExplained}=\text{FALSE}$) have $\text{ConfounderIdentity} \in \{\text{id-contested-org-choice, id-collider-proxy}\}$ OR $\text{CausalRole}=\text{contested}$ — no unexplained flip belongs to an unambiguous confounder identity.

Expected: $\text{UnexplainedConfounderSignFlipCount} = 0$

Observed: $\text{UnexplainedConfounderSignFlipCount}=0$

Corpus hypotheses — contingent patterns on this convenience sample. Directional inequalities only; not inferential. See conc-14.

H-age-identity-flip-rate — PASS (corpus hypothesis)

Age-composition identity ($\text{id-age-composition}$) produces manifest flip rate "e 40% among real studies — cross-domain demographic confounding is flip-prone, not latent-only.

Expected: $\text{AgeIdentityManifestFlipRate} \geq 0.40$ AND $\text{AgeIdentityStudyCount} \geq 10$

Observed: $\text{AgeIdentityManifestFlipRate}=0.486$ $\text{AgeIdentityStudyCount}=35$

H-catalog-exact-match — PASS (corpus hypothesis)

Catalog ExpectedDistortionType exact-match rate is below 50% — curator tags behave as mechanism priors, not taxonomy predictors.

Expected: $\text{TypePredictionMatchRate} < 0.5$

Observed: $\text{exactRate}=0.391$ (92/235)

H-causal-latent — PASS (corpus hypothesis)

Collider/selection studies produce latent-only flips under allocation sweep without manifest sign-flips at observed allocation.

Expected: $\text{ColliderSelectionManifestCount} = 0$ AND $\text{ColliderSelectionLatentOnlyCount} \geq 5$

Observed: $\text{collManifest}=0$ $\text{collLatent}=10$ $\text{collN}=11$

H-causal-manifest — PASS (corpus hypothesis)

Confounder-labeled studies produce manifest sign-flips at observed allocation ($\text{IsSignFlip}=\text{TRUE}$) far more often than collider/selection studies — reversal mode differs by mechanism class.

Expected: $\text{ConfounderSignFlipCount} > \text{ColliderSelectionManifestCount}$ AND $\text{ConfounderSignFlipCount} \geq 10$

Observed: $\text{confFlip}=85$ $\text{collManifest}=0$

H-collider-identity-low-manifest — PASS (corpus hypothesis)

Collider-proxy identity (id-collider-proxy) produces "d15% manifest flip rate — near-zero manifest, predominantly latent Type-D geometry.

Expected: $\text{ManifestFlipRate}(\text{cluster-id-collider-proxy}) \leq 0.15$ AND $\text{StudyCount}(\text{cluster-id-collider-proxy}) \geq 5$

Observed: $\text{ColliderIdentityManifestFlipRate}=0.091$ $\text{ColliderIdentityStudyCount}=11$

H-collider-no-manifest-v2 — PASS (corpus hypothesis)

Collider and selection studies produce zero manifest sign-flips (DistortionType A or B) at observed allocation — including hernandez-diaz birth-weight collider tagged Expected A in catalog.

Expected: $\text{ColliderSelectionManifestCount} = 0$

Observed: $\text{collManifest}=0$

H-confounder-drift-type-d — FAIL (corpus hypothesis)

When studies are bucketed by real-world valid-time (cohort midpoint, not publication year), case-mix / geographic / situational confounder archetypes (MechanismClass in severity-case-mix, geography, sports-situation) show $\text{DriftDirection}=\text{rising}$ toward clean Type-D outcomes in $\text{ConfounderDistortionTimeline}$, while age / institutional / SES archetypes (demographic-composition, organizational-unit, ses-income) show DriftDirection in (flat, single-decade) -- i.e. the drift is confounder-specific, not corpus-wide.

Expected: $\text{COUNTIFS}(\text{ConfounderDistortionTimeline}[\text{MechanismClass}], \text{"severity-case-mix|geography|sports-situation"}, \text{ConfounderDistortionTimeline}[\text{DriftDirection}], \text{"rising"}) = \text{COUNTIFS}(\text{ConfounderDistortionTimeline}[\text{MechanismClass}], \text{"severity-case-mix|geography|sports-situation"})$
AND $\text{COUNTIFS}(\text{ConfounderDistortionTimeline}[\text{MechanismClass}], \text{"demographic-composition|organizational-unit|ses-income"}, \text{ConfounderDistortionTimeline}[\text{DriftDirection}], \text{"rising"}) = 0$

Observed: severity=falling geography=flat situational=rising age=falling institutional=rising socioeconomic=rising — expected severity/geography/situational in (rising,single-decade) and age/institutional/socioeconomic $\neq$ rising; institutional and socioeconomic both drifted rising, contradicting the hypothesis.

H-cplus-magnitude — PASS (corpus hypothesis)

C+ studies carry higher mean AllocationDistortion than C- studies and Type-D studies — magnitude inflation is the largest quantitative error class and invisible to is_sign_flip.

Expected: CPlusAvgDistortion > CMinusAvgDistortion AND CPlusAvgDistortion > TypeDAvgDistortion

Observed: C+=0.111 C-=0.030 D=0.000

H-explained-confounder — PASS (corpus hypothesis)

IsParadoxExplained is TRUE for all and only confounder sign-flip studies — adjust-resolvable confounding is the unique condition that makes a reversal fully explained.

Expected: ExplainedConfounderCount = SignFlipCount AND ExplainedNonConfounderCount = 0

Observed: ExplainedConfounderCount=85 ConfounderSignFlipCount=85

H-geographic-stable-type-d — PASS (corpus hypothesis)

Geographic-composition identity (id-geographic-composition) has "e25% Type-D fraction — stable latent-D presence distinct from manifest-flip-prone identities.

Expected: TypeDFraction(cluster-id-geographic-composition) >= 0.25 AND StudyCount(cluster-id-geographic-composition) >= 3

Observed: GeographicTypeDFraction=0.647 GeographicStudyCount=17

H-latent-d — PASS (corpus hypothesis)

More than half of Type-D (SAFE) studies would flip their pooled winner under counterfactual allocation reweighting.

Expected: LatentTypeDFraction > 0.5

Observed: fraction=0.812

H-mechanism-other-drain — PASS (corpus hypothesis)

After ontology drain, id-mechanism-other contains "d5 studies — the catch-all is no longer the dominant bucket.

Expected: StudyCount(cluster-id-mechanism-other) <= 5

Observed: MechanismOtherStudyCount=0

H-purity-trap-flag-coverage — PASS (corpus hypothesis)

PurityTrapFlag=TRUE covers "e80% of collider/selection studies and "d5% of demographic-composition confounder studies — the flag correctly discriminates the trap from the safe-stratification zone.

Expected: PurityTrapCoverage(collider-selection) >= 0.80 AND PurityTrapFalsePositiveRate(demographic-composition) <= 0.05

Observed: ColliderSelectionCoverage=0.818 DemographicFalsePositiveRate=0.000

H-safety-corridor-lower-bound — PASS (corpus hypothesis)

The minimum observed MaxStratumImbalance among all manifest sign-flip studies is 0.033 (flu-vaccine-elderly-mortality). No sign-flip occurs below this threshold — the gap [0.0, 0.033] is a corpus-level safety corridor.

Expected: MinSignFlipStratumImbalance >= 0.033 AND SafeCorridorFlipCount=0

Observed: MinSignFlipStratumImbalance=0.033 SafeCorridorFlipCount=0

H-selection-frailty-zero-manifest — PASS (corpus hypothesis)

Selection-frailty identity (id-selection-frailty) produces zero manifest sign-flips — pure latent-D geometry, no allocation-reversal at observed weights.

Expected: ManifestFlipCount(cluster-id-selection-frailty) = 0 AND StudyCount(cluster-id-selection-frailty) >= 3

Observed: SelectionFrailtyManifestFlipCount=0 SelectionFrailtyStudyCount=5

H-severity-epi-latent-only — PASS (corpus hypothesis)

Disease-severity identity within epidemiology domain shows zero manifest flips — latent-only geometry in that domain.

Expected: ManifestFlipCount(cell-id-disease-severity-x-epidemiology) = 0

Observed: SeverityEpiManifestFlipCount=0

H-severity-medicine-manifest — PASS (corpus hypothesis)

Disease-severity identity within medicine domain has manifest flip rate "e45% — severity drives reversals in clinical settings.

Expected: ManifestFlipRate(cell-id-disease-severity-x-medicine) >= 0.45

Observed: SeverityMedicineManifestFlipRate=0.533

H-severity-vs-age-latent — FAIL (corpus hypothesis)

Disease-severity identity shows lower Type-D latent fraction than age-composition identity — severity drives manifest reversals more than hidden sweep sensitivity.

Expected: SeverityIdentityLatentFractionAmongTypeD < AgeIdentityLatentFractionAmongTypeD

Observed: SeverityIdentityLatentFractionAmongTypeD=0.000 AgeIdentityLatentFractionAmongTypeD=0.000

H-small-effect — PASS (corpus hypothesis)

Allocation-stable Type-D studies carry larger observed pooled gaps than latent Type-D studies.

Expected: AvgPooledGapStableD > AvgPooledGapLatentD

Observed: stable=0.113 latent=0.020

H-synthetic-edge-coverage — PASS (corpus hypothesis)

All four corpus theorem boundary regions (near-zero-distortion, equal-weight sign-flip, zero-corrected-gap, collider-purity) are covered by at least one synthetic study.

Expected: COUNTIF(Studies[IsSynthetic],TRUE) >= 4 AND all four boundary regions represented

Observed: SyntheticStudyCount=9 near-zero-distortion=t equal-weight-sign-flip=t zero-corrected-gap=t collider-purity=t

H-ultra-fragile — PASS (corpus hypothesis)

At least four Type-D studies are sweep-fragile: signal_purity=1, allocation_distortion=0, sweep_pooled_gap_range>0.3, allocation_fragility>=10 — perfectly safe by pooled metrics yet latent-flip under reweighting.

Expected: SweepFragileCount >= 4

Observed: SweepFragileCount=8

H-unexplained-nonconfounder — PASS (corpus hypothesis)

Studies with IsSignFlip=TRUE and CausalRole != confounder (contested or mediator) have IsParadoxExplained=FALSE.

Expected: ContestedOrMediatorExplainedCount = 0

Observed: ContestedOrMediatorExplainedCount=0

Vacuous corpus hypotheses — PASS only because their filtered universe is empty by construction (control studies were selected for absence of paradox and were never encoded with paired treatment arms, so there is no row that could sign-flip). True the way "all unicorns are blue" is true; excluded from the confirmation ratio above.

H-control-no-manifest-flip — PASS (vacuous)

Zero manifest sign flips among control studies. These studies were selected for absence of paradox, so no pooled-vs-stratified reversal should be present.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,Studies[IsControlStudy],TRUE) = 0

Observed: ControlManifestFlipCount=0 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-control-safety-corridor — PASS (vacuous)

No control study with MaxStratumImbalance < 0.033 (the safe-corridor threshold) has IsSignFlip=TRUE. The safety corridor hypothesis predicts zero flips below this boundary.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,TreatmentRankings[MaxStratumImbalance],"<0.033",Studies[IsControlStudy],TRUE) = 0

Observed: ControlSafeCorridorFlipCount=0 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-control-type-d-predominance — PASS (vacuous)

At least 70% of control studies encode as Type-D (neutral allocation, no sign flip) — Type-D is the expected basin for studies selected for absence of paradox.

Expected: $\text{COUNTIF}(\text{TreatmentRankings}[\text{IsSignFlip}], \text{TRUE}, \text{Studies}[\text{IsControlStudy}], \text{TRUE}) / \text{COUNTIF}(\text{Studies}[\text{IsControlStudy}], \text{TRUE}) < 0.3$

Observed: ControlFlipRate=0.000 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

Superseded corpus hypotheses (historical — retired from the live DAG at loop-79; not counted above). Six patterns witnessed FAIL at N=238 during expansion wave 3 (loop-78) and were removed from DiscoveryHypotheses so the instrument doesn't carry dead vocabulary forward — their FAIL evidence is preserved verbatim in conc-32-expansion-wave-3-supersession:

H-econ-zero — FAIL (superseded, loop-78/79): flips=10 (was 0 pre-expansion)

H-domain-dist — FAIL (superseded, loop-78/79): epi<edu (direction reversed)

H-catalog-flip-prediction — FAIL (superseded, loop-78/79): flipPred=54.1% (was 25.4%)

H-domain-flip-geometry-controlled — FAIL (superseded, loop-78/79): econHighImbFlips=10 (was 0)

H-econ-encoding-selection — FAIL (superseded, loop-78/79): mismatch=50% (was 100%)

H-domain-profiles-stable — FAIL (superseded, loop-78/79): econFlipRate=25.6% (was 0%)

Proved (by construction) (8)

conc-12-signal-purity-theorem · Proved (by construction)

SignalPurity < 0.5 is a necessary condition for allocation-driven reversal

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

Witnessed across all reversal studies (type A + B) in the 90+ study corpus: every study with AllocationDirection='reversal' has SignalPurity < 0.5. The algebraic proof: reversal requires SignedPooledGap and CorrectedGap to have opposite signs, which requires AllocationDistortion > |CorrectedGap|, which is exactly SignalPurity < 0.5. inv-signal-purity-sign-flip enforces this as a DAG invariant. AvgSignalPurityReversal "H 0.35 vs AvgSignalPurityNonReversal "H 0.65 — gap of ~0.30.

Loop replay: [loop-44](#)

Protecting invariants: 1

conc-20-signal-purity-ceiling · Proved (by construction)

SignalPurity < 0.5 is a build-breaking invariant for sign-flip studies — witnessed at corpus scale

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

At 96-study corpus scale, every manifest sign-flip has SignalPurity strictly below 0.5 (corpus max: 0.483, jeter-justice-1997). The melanoma-altman-1991 boundary case (purity=0.479, C+, pooled gap crosses zero, no flip) confirms the law is tight but not violated. This graduates from corpus hypothesis (loop-61) to build-breaking invariant: any future study encoding that produces a sign-flip with purity >= 0.5 is an encoding error, not a counterexample.

Loop replay: [loop-64](#)

conc-28-corrected-gap-invariance-theorem · Proved (by construction)

CorrectedGap (= WeightedStratumGapSum) is invariant under allocation reweighting — the conserved quantity

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

Loop-73: inv-corrected-gap-invariant PASS 238/238; MaxStudySweepCorrectedGapRange=0; CorrectedGapInvariantFailCount=0. CorrectedGap (=WeightedStratumGapSum) invariant under allocation reweighting across full corpus including expansion wave 3.

Loop replay: [loop-73](#)

Protecting invariants: 1

conc-29-explained-confounder-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

IsParadoxExplained 'ú confounder manifest sign-flip — adjust-resolvable confounding is the unique explanation condition

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-74: H-explained-bidirectional PASS. ExplainedConfounderCount=ConfounderSignFlipCount; FalsePositiveExplainedCount=0; UnexplainedConfounderSignFlipCount=0. inv-explained-sign-flip-confounder + inv-confounder-signflip-explained both PASS.

Loop replay: [loop-74](#)

Protecting invariants: 2

conc-30-collider-no-manifest-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Given correct causal-role annotation, collider/selection studies produce no manifest sign-flips at observed allocation

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-75: ColliderSelectionCount>=10 with ColliderSelectionManifestCount=0 after encoding heckman-1979 + hoxby-murarka-2009 and re-tagging simonsen/hammond to selection; fryer-2016 reclassified contested. inv-collider-no-manifest PASS.

Loop replay: [loop-75](#)

Protecting invariants: 1

conc-35-unexplained-identity-sink-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Identity-conditioned explained theorem: all unexplained sign-flips confined to contested-org-choice and collider-proxy identities

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-85 at N=238: H-unexplained-flips-only-nonconfounders PASS. UnexplainedConfounderSignFlipCount=0. The three unexplained flips (conc-29 exception list) map exclusively to id-contested-org-choice (2) and id-collider-proxy (1). This extends the loop-74 biconditional (conc-29): explained!"confounder holds for every unambiguous confounder identity; violations are confined to ontologically ambiguous categories. inv-unexplained-identity-sink: PASS (critical, 30 total invariants).

Loop replay: [loop-85](#)

Protecting invariants: 1

conc-36-perfect-balance-theorem · Proved (by construction)

Perfect allocation balance is a sufficient condition for no manifest Simpson's Paradox

Loop-86: inv-perfect-balance-no-flip PASS 118/118. All 118 TreatmentRankings with ArmSizeRatio=0.5 AND MaxStratumImbalance=0.0 have IsSignFlip=FALSE across all 8 domains and 19 confounder archetypes. Minimum imbalance at any sign-flip study: 0.033 (flu-vaccine-elderly-mortality, demographic-composition). Safety corridor [0.0, 0.033) contains zero sign-flips. This is a sufficient-condition theorem: perfect balance guarantees no manifest paradox, domain-agnostically.

Loop replay: [loop-86](#)

Protecting invariants: 1

conc-37-purity-inversion-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Purity inversion: high SignalPurity anti-predicts safe stratification — collider geometry maximizes it

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-87: collider/selection mean SignalPurity=0.916 (n=12); selection/frailty mean SignalPurity=1.000 (n=5) — the two mechanism classes that must NOT be stratified have the highest purity in the corpus. Demographic-composition (stratify-appropriate) mean SignalPurity=0.624 (n=35). inv-collider-purity-dominance PASS. H-purity-inversion confirmed. PurityTrapFlag encodes the trap for direct query. Researchers using data cleanliness as a quality heuristic will systematically misidentify collider/selection studies as...

Loop replay: [loop-87](#)
Protecting invariants: 1

Established in instrument (14)

conc-03-binary-excludes-partial · Established in instrument

The unanimity criterion (IsReversal) incorrectly excludes Berkeley 1973 despite its being a Simpson-type paradox

Berkeley: WSGSUM=" 0.0642, SignedPooledGap=+0.1292, IsSignFlip=TRUE, AllocationDistortion=0.1934 — greater distortion than kidney-1986. But IsReversal=FALSE because dept-C marginally favors male. The unanimity criterion is a strict subtype, not the primary definition.

Loop replay: [loop-17](#)
Protecting invariants: 5

conc-04-type-c-exists · Established in instrument

Allocation can compress the pooled signal without flipping it (type-C): a class invisible to IsReversal and IsSignFlip but visible to AllocationDistortion

compressed-synthetic: A wins every stratum and wins pooled, but the pooled margin is 6pp rather than the equal-weight 10pp. AllocationDistortion=0.0400, IsSignFlip=FALSE. The pooled winner is correct; the effect size is wrong by 4pp.

Loop replay: [loop-19](#)
Protecting invariants: 4

conc-05-real-data-generalizes · Established in instrument

Two real published studies (reintjes-2000, radelet-1981) hydrate as type-A full reversals without schema changes

Both studies entered via 05b-customize-data.sql. Witnessed: reintjes AllocationDistortion=0.0516, DistortionType=A; radelet AllocationDistortion=0.0684, DistortionType=A. No DAG or schema change was needed — the hydration contract generalizes.

Loop replay: [loop-20b](#)

conc-06-policy-derivable · Established in instrument

PolicyImplication is domain-neutral: the same formula produces researcher actions across medicine, law, and education

All five current studies (kidney-1986, berkeley-1973, compressed-synthetic, kidney-balanced, balanced-synthetic) plus two real studies produce unambiguous PolicyImplication values from DistortionType alone. No domain-specific knowledge is encoded in the formula.

Loop replay: [loop-21](#)

conc-07-model-self-portrait · Established in instrument

ModelSummary witnesses the instrument's own epistemic coverage as a first-class derived fact

ModelSummary aggregates ReversalCount, ExplainedCount, TypeA/B/C/D counts, and AvgAllocationDistortion across all studies. The model describes its own completeness in a single row.

Loop replay: [loop-10](#)

conc-11-reversal-recovery · Established in instrument

The model can produce the allocation-corrected winner directly from existing DAG fields — no new data required

CorrectedGap = WeightedStratumGapSum (already derived in loop-12/17). CorrectedWinner is a sign-read on CorrectedGap. CorrectedVsPooledAgreement is TRUE for Type-D studies and FALSE for Type-A/B studies. For kidney-1986: CorrectedGap=+0.0537, CorrectedWinner=A, CorrectedVsPooledAgreement=FALSE (pooled said B; corrected says A). The reversal recovery is free — it costs zero new data.

Loop replay: [loop-27](#)
Protecting invariants: 2

conc-15-latent-flip-potential · Established in instrument

Type-D at observed allocation does not imply allocation-stable — 70% of SAFE studies flip under counterfactual reweighting

Sensitivity over sweep range f'' [0.05, 0.95]; not a claim about realistic reallocation alone.

LatentTypeDCount=45, TypeDCount=64, LatentTypeDFraction=0.703. LatentFlipPotential=TRUE when DistortionType=D AND PooledGapCrossesZero=TRUE. Examples: uc-systemwide-admissions (Type D, sweep range 0.56), recsys-offline-eval (Type D, sweep range 0.56). The ScreeningTier SAFE label describes observed allocation only; sweep reveals latent flip potential.

Loop replay: [loop-61](#)

Protecting invariants: 1

conc-16-small-effect-fragile · Established in instrument

In this corpus, allocation-stable Type-D rows carry larger observed pooled gaps than latent Type-D rows

AvgPooledGapStableD=0.102 across 19 stable Type-D studies; AvgPooledGapLatentD=0.022 across 45 latent Type-D studies. Stable studies also have lower sweep range (avg 0.072 vs 0.211). Corpus-descriptive pattern on loop-61 convenience sample — not a deductive law. Small observed effects tend to be allocation-fragile even when currently classified SAFE.

Loop replay: [loop-61](#)

conc-18-causal-role-flip-mode · Established in instrument

Causal role predicts manifest vs latent reversal mode — confounders flip at observed allocation; collider/selection flips stay sweep-latent

Loop-62 research sweep: H-causal-manifest PASS (confFlip=15 vs collManifest=0). H-causal-latent PASS (collManifest=0, collLatent=5, collN=7). Confounder Type-D rows often show latent-only sweep sensitivity (45 latent-only) without manifest sign-flip — the mechanism split is about reversal mode when sign-flips occur, not absence of sweep fragility. Existing LatentFlipPotential + StratumCausalRole suffice; defer ManifestReversal first-class fields to loop-67+.

Loop replay: [loop-62](#)

conc-19-explained-tracks-confounding · Established in instrument

IsParadoxExplained tracks adjust-resolvable confounding, not geometric distortion

In this corpus, IsParadoxExplained=TRUE iff the study has a manifest sign-flip driven by a confounder — i.e. stratification resolves the reversal. It is FALSE for all C+/C-/D types, and FALSE for the three contested/mediator sign-flips (berkeley-1973, berkeley-six-dept-1973, birth-weight-paradox). The boundary separates 'stratify immediately' from 'do not naively adjust for a collider or mediator'.

Loop replay: [loop-63](#)

conc-24-collider-no-manifest-v2 · Established in instrument

Collider/selection studies produce zero manifest sign-flips at observed allocation — theorem candidacy

Loop-69: H-collider-no-manifest-v2 PASS. ColliderSelectionManifestCount=0 including hernandez-diaz birth-weight collider (Expected-A catalog tag, observed Type D). Catalog Expected-A does not predict collider geometry.

Loop replay: [loop-69](#)

conc-25-cplus-magnitude-blindspot · Established in instrument

Type-C+ carries the largest mean AllocationDistortion — invisible to is_sign_flip screens

Loop-69: H-cplus-magnitude PASS. CPlusAvgDistortion > CMinusAvgDistortion and > TypeDAvgDistortion. Magnitude inflation is the largest quantitative error class among non-flip distortions.

Loop replay: [loop-69](#)

conc-26-sweep-fragile-class · Established in instrument

Sweep-fragile Type-D class: pooled-safe studies with wide counterfactual wander ("e4 in corpus)

Loop-69: H-ultra-fragile PASS. SweepFragileCount>=4 — Type-D rows with signal_purity=1, allocation_distortion=0, sweep_pooled_gap_range>0.3, allocation_fragility>=10. Invisible to pooled-only screens.

Loop replay: [loop-69](#)

conc-33-confounder-identity-layer · Established in instrument

ConfounderIdentity ontology: cross-study accountable stratum archetypes (loop-80)

Loop-81 witness at N=238: 15 identities, 233/233 real stratum variables mapped (100% coverage). H-age-identity-flip-rate PASS (48.6% manifest, n=35). H-severity-vs-age-latent FAIL globally (0.917 > 0.813) — domain modulation deferred to loop-82 IdentityDomainCells.

Loop replay: [loop-81](#)

Taxonomy (1)

conc-13-cplus-cminus-opposite-errors · Taxonomy

C+ (amplification) and C- (compression) are epistemically opposite errors with opposite clinical implications

titanic-1912 (C+, DistortionRatio=1.359): allocation inflates first-class survival advantage from 27.5pp corrected to 37.4pp pooled — overconfidence. melanoma-altman-1991 (C+, DistortionRatio=2.087): allocation doubles apparent female survival advantage — severe overconfidence. hannan-1994 (C-, DistortionRatio=0.314): allocation compresses CABG advantage from 8.9pp corrected to 2.8pp pooled — underconfidence, risk of premature treatment abandonment. The prior type-C category hid this distinction. Now separated b...

Loop replay: [loop-43](#)

Methodology (7)

conc-10-instrument-specifiable · Methodology

The full instrument can be specified precisely enough for any researcher to apply it to a new study

Witnessed in loop-26: InstrumentSpec table enumerates 5 input fields, 10 derived coordinates, 2 classification rules, and 3 adapter-requirement rows. Machine-verifiable against the full 90+ study corpus.

Loop replay: [loop-26](#)

conc-14-scale-is-discovery-path · Methodology

Cross-domain SignalPurity correlations require a pre-registered unbiased corpus of 50+ studies to distinguish pattern from selection artifact

Partial domain patterns witnessed across 90+ studies (loop-61): epidemiology avg distortion 0.050 vs education 0.004; legal/sports/epidemiology flip rates ~27% vs economics 0%. Full causal domain!DistortionType mapping still requires pre-registered expansion beyond convenience sample, but the instrument now supports DiscoveryHypotheses/Findings for ongoing corpus-scale tests.

Loop replay: [loop-61](#)

conc-21-catalog-tags-not-predictors · Methodology

Catalog ExpectedDistortionType tags are mechanism priors, not taxonomy predictors — exact match rate 19.4%

Loop-65 audit: 18/93 exact matches (19.4%), 75 mismatches. Sign-flip prediction 18/71 (25.4%). Biggest bucket Expected A!observed D (23 studies). Both H-catalog hypotheses PASS (<50%). Defer ExpectedSignFlip vs ExpectedDistortionType split to loop-67+.

Loop replay: [loop-65](#)

conc-23-research-sweep-synthesis · Methodology

Research sweep loops 62–66: confirmed laws vs corpus patterns vs deferred vocabulary

Laws (build-breaking): SignalPurity<0.5 for sign-flips (loop-64), IsParadoxExplained!confounder (loop-63). Corpus patterns: causal-role reversal mode (loop-62), latent Type-D 70% (loop-61), catalog tag falsification 19% exact (loop-65), domain gap survives geometry control (loop-66). Deferred loop-67+: ManifestReversal/LatentReversal fields, ExpectedSignFlip catalog split, Discovery UI polish. inv-discovery-all-confirmed: 11/11 PASS.

Loop replay: [loop-66](#)

Protecting invariants: 1

conc-31-theorem-portfolio-synthesis · Methodology

Four witnessed theorems: SignalPurity, CorrectedGap invariance, Explained!"Confounder, Collider-no-manifest — plus corpus patterns that remain non-theorems

Loop-76: TheoremCount $\geq$ 4 (conc-12/20 SignalPurity + conc-28/29/30). All loops 61-75 discovery hypotheses PASS. Methods-export ready. **Loop-78 update:** six corpus patterns superseded at N=238; theorems and build-breaking invariants unchanged — see conc-32.

Loop replay: [loop-76](#)

Protecting invariants: 1

conc-32-expansion-wave-3-supersession · Methodology

Expansion wave 3 (loop-77!"78): six corpus patterns superseded at N=238 — theorems and invariants hold

Loop-78 LoopReview at 238 studies (233 real). Six EpistemicTier=corpus-pattern-superseded hypotheses witnessed FAIL: H-econ-zero (flips=10), H-domain-dist (epi<edu), H-catalog-flip-prediction (flipPred=54.1%), H-domain-flip-geometry-controlled (econHighImbFlips=10), H-econ-encoding-selection (mismatch=50%), H-domain-profiles-stable (econFlipRate=25.6%). H-catalog-exact-match still PASS (39.1%). Five theorems + all critical invariants unchanged. Corpus patterns are scale- and selection-contingent; laws are not.

Loop replay: [loop-78](#)

conc-expansion-wave-synthesis · Methodology

Expansion waves 1–2 (loops 68–72): adversarial stress tests survived; theorem candidacy patterns held or sharpened

Pending loop-72 LoopReview. Witnesses collider-no-manifest, economics encoding selection, sweep-fragile class, domain profile stability at N>110.

Loop replay: [loop-76](#)

Scope / open question (2)

conc-08-type-c-real · Scope / open question

Does the published literature contain real type-C studies (compression without sign flip)?

Witnessed in loop-24: hannan-1994 (CABG surgery), titanic-1912, melanoma-altman-1991, and coffee-tverdal-2020 all hydrate as DistortionType=C. AllocationDistortion>0.01, IsSignFlip=FALSE — the correct pooled direction but compressed magnitude. The type-C cell is not a synthetic pathology.

Loop replay: [loop-24](#)

conc-09-causal-vs-geometric · Scope / open question

AllocationDistortion is a geometric measure; causal interpretation requires external domain knowledge, gated by AdjustmentAppropriate

Berkeley has the highest AllocationDistortion (0.1934) but CausalRole=contested (mediator vs confounder), proving that geometric severity and causal confounding are not the same thing. The instrument is deliberately geometric: it classifies distortion type and flags the corrected winner from allocation arithmetic alone. Whether it is safe to act on that correction as a causal claim is answered by AdjustmentAppropriate — which gates on ConditioningRisk and CausalClaimStatus. The two layers are complementary, not ...

Loop replay: [loop-30](#)

Protecting invariants: 1

Observed in this corpus (2)

conc-01-paradox-is-derived · Observed in this corpus

Simpson's Paradox is a derived fact from the DAG, not a modeled entity

IsReversal (and later IsSignFlip) falls out as a calculated field on TreatmentRankings, derived from aggregated CaseCell counts. No 'ReversalDetection' entity was ever needed.

Loop replay: [loop-04](#)
Protecting invariants: 2

conc-02-reversal-from-allocation · Observed in this corpus

The reversal in kidney-1986 is entirely an allocation artifact — same rates with balanced allocation produce no reversal

kidney-balanced uses identical stratum success rates to kidney-1986 but with equal allocation (175 cases per cell).
IsReversal=FALSE, AllocationDistortion=0. The paradox is algebraically isolated as a property of allocation, not of treatment efficacy.

Loop replay: [loop-14](#)
Protecting invariants: 3

Historical (superseded — no longer live) (3)

These conclusions were true of an earlier, smaller corpus and were superseded by later expansion waves — kept visible with their supersession evidence rather than deleted.

conc-17-economics-flip-free · corpus pattern superseded

Economics-domain 2×K encodings in this corpus produce zero sign-flips — composition studies read as Type D or C"

This corpus only (n=8 economics studies); see conc-14 for inferential limits.

EconomicsSignFlipCount=0 across 8 economics studies (6 Type D, 2 Type C"). Contrast epidemiology (6 sign-flips / 22 studies, 27.3%). Composition and wage-decomposition paradoxes compress or stabilize rather than flip in this instrument's 2×K encoding.
Superseded at loop-78 (N=238): EconomicsSignFlipCount=10 (25.6% flip rate) after expansion wave 3 bulk encode — pattern was corpus-contingent on pre-expansion selection, not a domain law.

Loop replay: [loop-61](#)

conc-22-domain-flip-geometry · corpus pattern superseded

Domain flip-rate heterogeneity survives max_stratum_imbalance conditioning — economics 0%, epi/legal/sports ~27%

Corpus-specific pattern; see conc-14.

Loop-66: H-domain-flip-geometry-controlled PASS. economics_high_imbalance_sign_flip_count=0; epidemiology_high_imbalance_sign_flip_rate=0.273; legal=0.286; sports=0.273. Pattern survives geometry control — not purely catalog selection artifact. **Superseded at loop-78 (N=238):** econHighImbFlips=10; all domains elevated (epi=65%, legal=57%, sports=60%) — geometry-controlled domain gap collapsed after wave-3 encode.

Loop replay: [loop-66](#)

conc-27-economics-encoding-selection · corpus pattern superseded

Economics Expected-A catalog tags mismatch observed Type D at 2×K encoding — selection not immunity

Loop-69: H-econ-encoding-selection PASS. EconomicsExpectedAMismatchRate=1.0 on expansion-wave-1 Expected-A batch (5/5 Type D). Economics zero-flip reflects encoding limits and catalog priors, not domain paradox immunity. **Superseded at loop-78 (N=238):** EconomicsExpectedAMismatchRate=0.50 after wave-3 dilution — half of Expected-A economics rows now match; pattern sharpened not falsified but no longer unanimous.

Loop replay: [loop-69](#)

Invariant health

Critical invariant failures: [0](#)

Definitional (cannot fail by construction — see definitions above): 22

Judgment-crossing (touch hand/LLM-encoded CausalRole): 4

Vacuous (0 pass / 0 fail — empty universe, proves nothing yet): 0

All 26 critical invariants pass — 22 are transpiler regression checks against stated algebra (cannot fail by construction), 4 cross into the hand/LLM-encoded CausalRole field, and 0 are currently vacuous (empty universe). See "How to read "InvariantChecks" (read

before the pass count)" above.

Warning-only (informational, not critical — pre-loop-70 legacy checks kept for visibility, do not gate the build):

inv-import-session-ready: Loop-67+: expansion backlog loaded; bulk encode session ready when ≥ 20 candidates and ≥ 15 encode-ready. (fail=1)

inv-pooled-gap-crosses-zero: Legacy sweep check from ratio-based Type A (pre-loop-70). Type A is now unanimity-based; cross-zero is informative but not definitional. (fail=1)

inv-pooled-gap-wanders: Pooled gap should move under allocation reweighting for reversal studies. Warning-only: small-magnitude reversals may have narrow sweep range. (fail=1)

inv-type-a-ratio-negative: Legacy ratio geometry check (pre-loop-70). Type A is now IsStratumUnanimous; ratio band retained as warning-only witness. (fail=4)

inv-type-b-ratio-lt-neg1: Legacy ratio geometry check (pre-loop-70). Type B is now heterogeneous stratum directions; ratio threshold retained as warning-only witness. (fail=2)

Judgment-crossing invariants (the closest thing here to a substantive check):

inv-collider-no-manifest: Real-data collider and selection studies never produce manifest sign-flips at observed allocation — conditional theorem requiring correct causal-role annotation (loop-75). Scoped to real-data studies: loop-89's collider-purity-boundary-flip is a deliberately-constructed synthetic adversarial stress test (1.2M-trial search for the highest-purity manifest flip a collider geometry can produce, purity=0.49994) built specifically to probe this theorem's boundary from the outside — it is evidence the boundary is real and tight, not a counterexample to the real-data claim (loop-92). (pass=11, fail=0)

inv-confounder-signflip-explained: Every confounder manifest sign-flip must be marked explained. Other half of the explained!"confounder biconditional (loop-74). (pass=85, fail=0)

inv-explained-sign-flip-confounder: Every IsParadoxExplained=TRUE row must be a confounder manifest sign-flip. Half of the explained!"confounder biconditional (loop-74). (pass=85, fail=0)

inv-unexplained-identity-sink: Every real-data unexplained sign-flip must be linked to contested-org-choice or collider-proxy identity — no unexplained flip belongs to an unambiguous confounder archetype. Identity-conditioned extension of conc-29 (loop-74 biconditional). Scoped to real-data studies (loop-92): loop-89's collider-purity-boundary-flip is a deliberately-constructed synthetic study with no StratumVariableIdentityMaps row at all (synthetic studies are never identity-mapped), so it is not a counterexample to this real-corpus claim. (pass=3, fail=0)

Most misleading studies (by paradox strength)

COVID-19 CFR: Italy vs China by Age (von Kügelgen 2021) — type C+, strength 0.2743, distortion 0.529, purity 3.6% (CAUTION)

Triathlon On-Time Finish: Sex by Age Cohort — type A, strength 0.1455, distortion 0.245, purity 28.9% (DANGER)

Smoking and Mortality by Age (Cochran / Surgeon General 1964) — type A, strength 0.1091, distortion 0.167, purity 25.7% (DANGER)

UC Berkeley Graduate Admissions (Bickel et al. 1975) — contested confounder — type B, strength 0.0969, distortion 0.193, purity 24.9% (DANGER)

UC Berkeley Admissions: Six Largest Departments (Bickel et al. 1975) — type B, strength 0.0944, distortion 0.184, purity 18.8% (DANGER)

Exercise and Cholesterol Control by Age Group (Framingham-style) — type D, strength 0.0800, distortion 0.008, purity 95.2% (SAFE)

Baseball Batting Averages: Lefebvre vs Fairly World Series (Day 1994 / baseball paradox canon) — type A, strength 0.0606, distortion 0.082, purity 20.9% (DANGER)

Whickham Smoking Cohort — 20-year Mortality by Smoking Status (Appleton et al. 1996) — type B, strength 0.0538, distortion 0.119, purity 26.9% (DANGER)

Rulebook & repo (full derivation)

Formulas, Loops build history, InvariantChecks, and Conclusions rows live in the rulebook JSON. README.md carries the full framing.

Project folder: <https://github.com/EffortlessAPI/effortless-rulebooks/tree/main/rulebook-examples/simpsons-paradox>

README (full repo guide): [README.md](#)

Rulebook SSoT (formulas, Loops, InvariantChecks, Conclusions): [effortless-rulebook/simpsons-paradox-rulebook.json](#)

Rulespeak (plain-English business rules): [rulespeak/rulespeak.md](#)

Postgres substrate (views, init-db.sh): [effortless-postgres/README.md](#)

Standalone HTML explorer: [simpsons-paradox-explorer.html](#)

Live API (with ./start.sh): GET localhost:3001/api/invariant-checks · /api/treatment-rankings · /api/model-summary